CS 105: Department Introductory Course on Discrete Structures

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Lecture 16 – Counting and Combinatorics

Logistics

Quiz 2: Oct 23 Thursday 3.45pm

- Syllabus: Counting and Combinatorics, bit of Graph theory
- Venue: To be announced

Pop Quiz: Anytime

Counting and Combinatorics

Topics

- Basics of counting
  - Product principle
  - Sum principle
  - Bijection principle
  - Double counting
- Subsets, partitions, Permutations and combinations
  1. Binomial coefficients and Binomial theorem
  2. Pascal's triangle
  3. Permutations and combinations with repetitions
  4. Estimating $n!$
- Recurrence relations and generating functions
- Today: Principle of Inclusion and Exclusion and its applications
- Today Pigeonhole Principle and its extensions

Principle of Inclusion-Exclusion (PIE)

A simple example:

- If in a class n students like python, m students like C and k students who like both, and ℓ like neither, then how many students are there in the class?
- Of course, this also counts the no. who were too lazy to lift their hands!

Theorem: Principle of Inclusion-Exclusion (PIE)

Let $A_1, A_2, \ldots, A_n$ be finite sets. Then,

$$|A_1 \cup \ldots \cup A_n| = \sum_{1 \le i \le n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j|$$

$$+ \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| - \ldots + (-1)^{n+1} |A_1 \cap \ldots \cap A_n|$$

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Proof: (H.W): Prove PIE by induction.

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Proof:

- We will show that each element in the union is counted exactly once in r.h.s
- Suppose a belongs to exactly r of the sets $A_1, \ldots, A_n$.
- Then a is counted $\binom{r}{1}$ times by $\sum |A_i|$, etc.
- Thus, overall count $= \binom{r}{1} - \binom{r}{2} + \ldots (-1)^{r+1} \binom{r}{r}$.
- What is this number?! =1!

Application: Number of surjections

- How many surjections are there from $[n] = \{1, \ldots, n\}$ to $[m] = \{1, \ldots, m\}$?
- # surjections = total #functions - those that miss some element in range.
- Let $A_i = \{f : [n] \to [m] \mid i \notin Range(f)\}$
- Then, # surjections $= m^n - |\cup_{i \in [m]} A_i|$.
- $|\cup_{i \in [m]} A_i| = \sum_{1 \le i \le m} |A_i| - \sum_{1 \le i < j \le m} |A_i \cap A_j| + \sum_{1 \le i < j < k \le m} |A_i \cap A_j \cap A_k| - \ldots + (-1)^{m+1} |A_1 \cap \ldots \cap A_m|$
- But now what is $|A_i|, |A_i \cap A_j|, |A_i \cap A_j \cap A_k|, \ldots$?
- $|A_i| = (m-1)^n$, $|A_i \cap A_j| = (m-2)^n$...
- What about the summation? terms $1 \le i < j \le m = \binom{m}{2}$

Thus, we have # surjections from $[n]$ to $[m]$ =

$$m^n - \binom{m}{1}(m-1)^n + \binom{m}{2}(m-2)^n - \ldots + (-1)^{m-1}\binom{m}{m-1} \cdot 1^n.$$

Pop Quiz!

1. Does there exist an injective function from a set of $k+1$ elements to a set with k elements? Why or why not?
2. How many cards must be selected from a pack of 52 cards so that at least three cards of the same suit are chosen?
3. Prove or disprove
   - 3.1 For every $n \in \mathbb{Z}^+$, there exists a multiple of n whose decimal expansion only has $0's$ and $1's$.
   - 3.2 Every sequence of n^2+1 distinct real numbers contains a subsequence of length $n+1$ which is either increasing or decreasing.
   - 3.3 If there are $n \geq 1 + r(\ell - 1)$ objects which are colored with r different colors, then there exist ℓ objects all with the same color.

This lecture

Pigeon-Hole Principle (PHP) and its applications

A simple formulation

Let $k \in \mathbb{N}$. If $k+1$ (or more) objects are to be placed in k boxes, then at least one box will have 2 (or more) objects.

How do you prove it?
A simple corollary

- Can a function from a set of $k+1$ or more elements to a set with k elements be injective?

Pigeon-Hole Principle (PHP)

Another simple application

For every $n \in \mathbb{Z}^+$, there exists a multiple of n whose decimal expansion only has $0's$ and $1's$.

- Consider $n+1$ integers $k_1 = 1$, $k_2 = 11$, $k_3 = 111$, $\ldots$, $k_{n+1} = 1\ldots1$ (with $n+1$ 1's).
- When any integer is divided by n, the remainder can be either $0, 1, \ldots, n-1$, i.e., n choices.
- So among the $n+1$ integers, by PHP, at least 2 must have the same remainder.
- That is, $\exists i, j$, $k_i = pn + d$, $k_j = qn + d$.
- But then $|k_i - k_j|$ is a multiple of n and its decimal expansion only has $0's$ and $1's$. $\square$

A (slightly) more general PHP

Pigeon-Hole Principle (Variant 1)

If N objects are placed into k boxes then there is at least one box with at least $\lceil N/k \rceil$ objects.

Suppose not. Then each box has strictly less than $\lfloor N/k \rfloor$ objects. Therefore, totally there can be strictly less than N objects, which is a contradiction.

A simple example

How many cards must be selected from a pack of 52 cards so that at least three cards of the same suit are chosen?

- 4 boxes, one for each suit; each selected card is put in one.
- If N cards are selected then at least 1 box has $\lceil N/4 \rceil$ cards.
- To have ≥ 3 cards from same suit, suffices $\lceil N/4 \rceil \geq 3$.
- Thus, $N = 9$. But can we do better? No. $\square$

Another application of PHP

Question: Give a sequence of 10 real numbers with no subsequence of length 4 which is increasing or decreasing.

Theorem

Every sequence of $n^2 + 1$ distinct real numbers contains a subsequence of length $n + 1$ which is either increasing or decreasing.

Applications

- In sorting algorithms, bioinformatics: largest block of genes with same order in genome of two species (called syntheny blocks!)
- Ranking of players: longest monotone subsequence is the largest chain of players ordered consistently.

Another application of PHP

Theorem

Every sequence of $n^2 + 1$ distinct real numbers contains a subsequence of length $n + 1$ which is either increasing or decreasing.

1. Let $a_1, \ldots, a_{n^2+1}$ be a sequence of distinct real numbers.
2. For each $k \in \{1 \ldots n^2 + 1\}$, let (i_k, d_k) denote a pair:

$$\begin{aligned} i_k &= \text{length of longest increasing subsequence starting from } a_k \\ d_k &= \text{length of longest decreasing subsequence starting from } a_k \end{aligned}$$

3. Suppose, there are no increasing/decreasing subsequences of length $n + 1$. Then $\forall k$, $i_k \leq n$ and $d_k \leq n$.
4. $\therefore$ by PHP, $\exists \ell, m, 1 \leq \ell < m \leq n^2 + 1$ s.t. $(i_\ell, d_\ell) = (i_m, d_m)$
5. We will show that this is not possible: 2 cases possible.
   - Case 1: $a_\ell < a_m$. Then $i_m \geq i_\ell + 1$, a contradiction.
   - Case 2: $a_\ell > a_m$. Then $d_\ell \geq d_m + 1$, a contradiction.
6. All a_i's are distinct so this completes the proof. $\square$

Another variant of PHP

Theorem (PHP Variant 2)

Suppose there are $n \geq 1 + r(\ell - 1)$ objects which are colored with r different colors. Then there exist ℓ objects all with the same color.

Proof: (H.W)

- Is this coloring optimal?
- That is, if fewer than $1 + r(\ell - 1)$ objects are given, is there a way of coloring them such that no ℓ have the same color?

The coloring game

- There are six points on board and two colored chalks.
- Divide class into 2 groups. Group 1 draws white lines and Group 2 draws blue lines between points.
- You lose if you are first to draw a triangle of your color.
- Can you ever have a draw?

We will now show that this is impossible. That is,

Lemma

Any 2-coloring of edges of a graph on 6 nodes has a monochromatic triangle.

- 2-coloring of edges: coloring all edges of the graph using atmost 2 colors.
- monochromatic (triangle): all edges have the same color.